The sum and direct sum of subspaces

S_1 and S_2 are subsets of vector space V

Sum: $S_1 S_2$ nonempty

$S_1 + S_2 = \{x + y \mid x \in S_1 \text{ and } y \in S_2\}$

Direct sum: $S_1 \cap S_2 = \{0\}$ and $S_1 + S_2 = V$

$\rightarrow V = S_1 \oplus S_2$

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29. **(a)** Prove that if W_1 and W_2 are finite-dimensional subspaces of a vector space V, then the subspace $W_1 + W_2$ is finite-dimensional, and $\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$. *Hint:* Start with a basis $\{u_1, u_2, \ldots, u_k\}$ for $W_1 \cap W_2$ and extend this set to a basis $\{u_1, u_2, \ldots, u_k, v_1, v_2, \ldots v_m\}$ for W_1 and to a basis $\{u_1, u_2, \ldots, u_k, w_1, w_2, \ldots w_p\}$ for W_2.

(a) If we assume

$$\sum_{i=1}^{k} a_i u_i + \sum_{i=1}^{m} b_i v_i + \sum_{i=1}^{p} c_i w_i = 0$$

Then

$$v = \sum_{i=1}^{m} b_i v_i = -\sum_{i=1}^{k} a_i u_i - \sum_{i=1}^{p} c_i w_i$$

$\in W_1$ $\quad$ $\in W_2$

$\rightarrow v \in W_1 \cap W_2 \rightarrow$ if $v \neq 0$ $\quad v = \sum_{i=1}^{k} a_i' u_i$

$\rightarrow \sum_{i=1}^{m} b_i v_i - \sum_{i=1}^{k} a_i' u_i = 0 \rightarrow$ contradictory

Thus $v = \sum_{i=1}^{m} b_i v_i = -\sum_{i=1}^{k} a_i u_i - \sum_{i=1}^{p} c_i w_i = 0$

$\rightarrow a_i = b_j = c_l = 0$ for all i, j, k

$\rightarrow \beta = \{u_1, \cdots, v_1 \cdots, w_1, \cdots\}$

is linearly independent

For $x \in W_1$, $y \in W_2$

$x = \sum_{i=1}^{k} a_i u_i + \sum_{i=1}^{m} b_i v_i$

$y = \sum_{i=1}^{k} a_i' u_i + \sum_{i=1}^{n} c_i w_i$

Hence

$x + y = \sum_{i=1}^{k} (a_i + a_i') u_i + \sum_{i=1}^{m} b_i v_i + \sum_{i=1}^{n} c_i w_i$

is a linear combination of β

$\rightarrow \dim(W_1 + W_2) = k + m + n$

$= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$

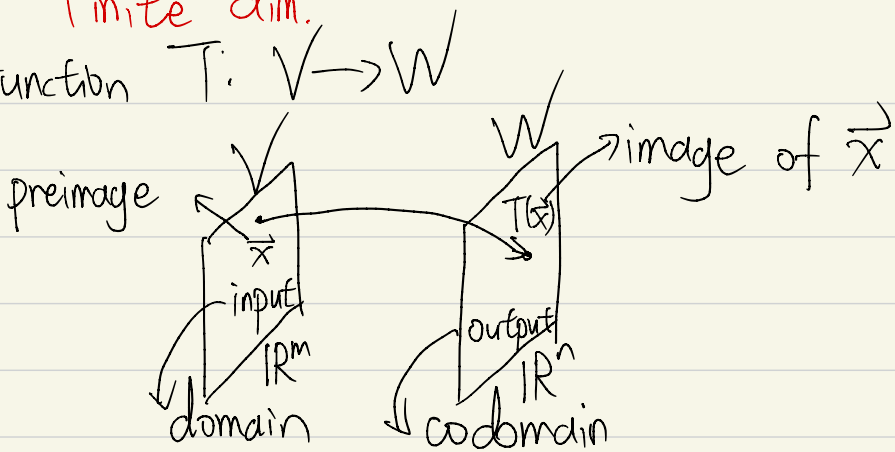
(b) Let W_1 and W_2 be finite-dimensional subspaces of a vector space V, and let $V = W_1 + W_2$. Deduce that V is the direct sum of W_1 and W_2 if and only if $\dim(V) = \dim(W_1) + \dim(W_2)$.

$\leftrightarrow \dim(W_1 \cap W_2) = 0 \leftrightarrow W_1 \cap W_2 = \{0\}$

$\leftrightarrow V = W_1 \oplus W_2$

Finite dim.

Function $T: V \to W$



Linear transformation

(a) $T(x+y) = T(x) + T(y)$

(b) $T(cx) = cT(x)$

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(a) If T is linear, then T preserves sums and scalar products. ✓

(b) If $T(x+y) = T(x) + T(y)$, then T is linear. ✗

(d) If T is linear, then $T(0_V) = 0_W$. ✓ $T(0x) = 0T(x) = 0_W$

(f) If T is linear, then T carries linearly independent subsets of V onto linearly independent subsets of W. ✗ map to 0

(g) If $T, U: V \to W$ are both linear and agree on a basis for V, then $T = U$. ✓

(h) Given $x_1, x_2 \in V$ and $y_1, y_2 \in W$, there exists a linear transformation $T: V \to W$ such that $T(x_1) = y_1$ and $T(x_2) = y_2$. ✗

(g) $\alpha = \{v_1, \cdots, v_m\}$ is a basis for V ✓

For arbitrary vectors $v = \sum_{i=1}^{m} a_i v_i$

$$T(v) = T\left(\sum_{i=1}^{m} a_i v_i\right) = \sum_{i=1}^{m} T(a_i v_i) = \sum_{i=1}^{m} a_i T(v_i)$$

$$= \sum_{i=1}^{m} a_i U(v_i) = U(v)$$

(h) $x_2 = 2x_1 \longrightarrow y_2 = T(x_2) = 2T(x_1) = 2y_1$

Range and null space

$T: V \to W$ linear

	def.	subspace of	dim.
range	$R(T) = \{T(\vec{x}) : \vec{x} \in V\}$	W	$\text{rank}(T)$
null space	$N(T) = \{\vec{x} \in V : T(\vec{x}) = 0\}$	V	$\text{nullity}(T)$

β is a basis for $V \to R(T) = \text{span}(T(\beta))$

Injective and surjective

general functions

injective (one to one): for $f(x)$ there exist a unique preimage

surjective (onto): range = codomain

linear: $\leftrightarrow N(T) = \{0\}$ $\leftrightarrow$ nullity $= 0$

linear: $\leftrightarrow \text{rank}(T) = \dim(W)$

bijective

(c) T is one-to-one if and only if the only vector x such that $T(x) = 0$ is $x = 0$. ✗

T is linear: ✓ $\to$: $f(x) = x + 1$

$\leftarrow$: $f(x) = |x|$

Dimension theorem

$$\text{nullity}(T) + \text{rank}(T) = \dim(V)$$

For Exercises 2 through 6, prove that T is a linear transformation, and find bases for both N(T) and R(T). Then compute the nullity and rank of T, and verify the dimension theorem. Finally, use the appropriate theorems in this section to determine whether T is one-to-one or onto.

5. $T: P_2(R) \to P_3(R)$ defined by $T(f(x)) = xf(x) + f'(x)$.

$P_n(R) = \sum_{i=0}^{n} a_i x^i$ $f(x) = a_0 + a_1 x + a_2 x^2$ $\dim(P_2) = 3$

$$T(f(x)) + T(g(x)) = xf(x) + f'(x) + xg(x) + g'(x)$$
$$= x[f(x) + g(x)] + [f(x) + g(x)]'$$
$$= T(f(x) + g(x))$$

$$T(cf(x)) = cxf(x) + cf'(x) = cT(f(x))$$

$\longrightarrow$ T is linear

$xf(x) + f'(x) = 0$

$\longrightarrow x(a_0 + a_1 x + a_2 x^2) + a_1 + 2a_2 x = 0$

$\longrightarrow N(T) = \{0\}$ with basis ϕ nullity $= 0$

$R(T) = \{a_2 x^3 + a_1 x^2 + (a_0 + 2a_2)x + a_1\}$

$= \{a_2 x^3 + a_1(x^2 + 1) + (a_0 + 2a_2)x\}$

with basis $\{x^3, x^2 + 1, x\}$ rank $= 3$

one-to-one but not onto

6. $\mathrm{T}: \mathrm{M}_{n\times n}(F) \to F$ defined by $\mathrm{T}(A) = \mathrm{tr}(A)$.

$tr(A) = \sum_{i=1}^{n} A_{ii}$ linear $\dim(M_{n\times n}) = n^2$

$N(T)$: $tr(A) = 0$

basis $\{E_{ij}\}_{i\neq j} \cup \{E_{ii} - E_{nn}\}_{i=1,2,\cdots,n-1}$

nullity$(T) = n^2 - n + n - 1 = (n+1)(n-1) = n^2 - 1$

$R(T) = F$ with basis 1 rank$(T) = 1$

onto but not one-to one

1. Let $a_1, a_2, \cdots, a_m$ $(\mathrm{m} \geq 2)$ be vectors in a vector space over $\mathbb{R}$. a_m is not a zero vector. Prove that for any real number $k_1, k_2, \cdots, k_{m-1}$, a set $S = \{\beta_1, \beta_2, \cdots, \beta_{m-1}\}$ $(\beta_1 = a_1 + k_1 a_m, \beta_2 = a_2 + k_2 a_m, \cdots, \beta_{m-1} = a_{m-1} + k_{m-1} a_m)$ is linearly independent if and only if $\{a_1, a_2, \cdots, a_m\}$ is linearly independent. (16 pts)

$\Leftarrow$: $\sum_{i=1}^{m} c_i d_i = 0 \to$ all $c_i = 0$

For $\sum_{i=1}^{m-1} c_i \beta_i = \sum_{i=1}^{m-1} (c_i d_i + c_i k_i d_m)$

$= \sum_{i=1}^{m-1} c_i d_i + \left(\sum_{i=1}^{m-1} c_i k_i\right) d_m = 0$

$c_i = 0$, $\sum_{i=1}^{m-1} c_i k_i = 0$

$\to \{\beta_i\}$ is linearly independent

$\Rightarrow$: $\sum_{i=1}^{m-1} c_i\beta_i = \sum_{i=1}^{m-1} c_i d_i + \left(\sum_{i=1}^{m-1} c_i k_i\right) d_m = 0$

$\rightarrow$ all $c_i = 0$, $i = 1, \cdots, m-1$

If $\{d_1, \cdots, d_m\}$ is linear dependent,

there exist $\sum_{i=1}^{m} s_i d_i = 0$, where there's nonzero s_i

Take $c_i = s_i$, $i = 1, \cdots, m-1$,

and take $\{k_i\}$ s.t. $\sum_{i=1}^{m-1} c_i k_i = s_m$ $\rightarrow$ contradiction

【例 6.123】(南开大学,2009 年) 设 P 为数域,T 为 $P^{n\times n}$ 上的线性变换,满足条件:对任何固定的 $\boldsymbol{A},\boldsymbol{B}\in P^{n\times n}$,$T(\boldsymbol{AB})=T(\boldsymbol{A})T(\boldsymbol{B})$ 或 $T(\boldsymbol{AB})=T(\boldsymbol{B})T(\boldsymbol{A})$ 至少有一个成立. 证明:要么对所有的 $\boldsymbol{A},\boldsymbol{B}\in P^{n\times n}$,$T(\boldsymbol{AB})=T(\boldsymbol{A})T(\boldsymbol{B})$,要么对所有的 $\boldsymbol{A},\boldsymbol{B}\in P^{n\times n}$,$T(\boldsymbol{AB})=T(\boldsymbol{B})T(\boldsymbol{A})$.

Theorem: If a vector space is the union of two subspaces, then one of them equals the whole space

Proof. if $W = U \cup V$ and both are proper subspaces

choose $u \in U$, $v \in V$. Consider $u+v$

If $u+v \in V \rightarrow v \in U \rightarrow \times$

If $u+v \in V \rightarrow u \in V \rightarrow \times$

$u+v$ belongs to neither subspace

$\rightarrow$ a contradiction

Now write $M = F^{n\times n}$. For $A \in M$

define $U_A = \{B \in M : T(AB) = T(A)T(B)\}$

$V_A = \{B \in M : T(AB) = T(B)T(A)\}$

both are subspaces of M

The hypothesis gives

$$M = U_A \cup V_A$$

Thus $U_A = M$ or $V_A = M$

Next, define

$U = \{A \in M : T(AB) = T(A)T(B)$ for every $B \in M\}$

$V = \{A \in M : T(AB) = T(B)T(A)$ for every $B \in M\}$

$$M = U \cup V$$

$\rightarrow U = M$ or $V = M$

Hence either

$$T(AB) = T(A)T(B) \text{ for all } A, B \in M$$

or

$$T(AB) = T(B)T(A) \text{ for all } A, B \in M$$